

Bidirectional Transport along Microtubules

Review

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Active transport by microtubule motors has a plethora of crucial roles in eukaryotic cells. Organelles often move bidirectionally, employing both plus-end and minus-end directed motors. Bidirectional motion is widespread and may allow dynamic regulation, error correction and the establishment of polarized organelle distributions. Emerging evidence suggests that motors for both directions are simultaneously present on cellular ‘cargo’, but that their activity is coordinated so that when plus-end motors are active, minus-end motors are not, and *vice versa*. Both the dynein cofactor dynactin and the Karsicht (Klar) protein appear to be important for such coordination. The direction of net transport depends on the balance between plus-end directed and minus-end directed motion. In several model systems, factors crucial for setting this balance have now been identified, setting the stage for a molecular dissection of the underlying regulatory mechanisms. These analyses will likely provide insight into motor cooperation in general.

Introduction

All cells are highly organized spatially. To help establish and maintain uneven distributions of specific proteins, RNAs and organelles, eukaryotic cells employ several distinct mechanisms for active large-scale transport, including actin polymerization, cytoplasmic streaming and vectorial movement along cytoskeletal fibers. Of these mechanisms, motor-driven transport of cargo along microtubules and microfilaments is the best understood. These motors are highly regulated *in vivo* so that cargoes are transported to distinct locations, often under precise spatial and temporal control. How cells achieve specificity, directionality and timing of transport is poorly understood. The docking of motors to their cargoes is potentially a key control point, and consequently the search for receptors and adaptors that link motors to cargoes has been a major focus of research. This search has yielded surprising biological connections; for example, one of the receptors for the motor kinesin I is the amyloid precursor protein [1,2], a finding that implicates kinesin-based axonal transport in the development of Alzheimer’s disease [3].

Despite these exciting advances, many aspects of motor regulation are barely understood. In particular, there is a whole class of transport processes that employ multiple, distinct motors that act either simultaneously or alternately in quick succession. Motor

cooperation gives rise to emergent phenomena that cannot be easily accounted for by the properties of individual motors or by cargo docking, raising entirely new issues about motor function and regulation. This review will focus on the conceptually simplest case, transport by microtubule motors of opposing directionality, and will develop its distinct features and biological significance (Box 1). Because bidirectional transport is widespread, these issues are important for a broad range of biological phenomena, including the establishment of cell polarity during development, trafficking of intracellular HIV particles and neurodegenerative diseases.

Two Distinct Ways to Move

Microtubules are polarized, with chemically and morphologically distinct plus and minus ends. In a typical eukaryotic cell, microtubules are arranged with their minus ends clustered around a microtubule organizing center near the nucleus and their plus ends pointing towards the periphery, though many exceptions exist. Microtubule motors recognize the inherent polarity of these tracks, and most motors move exclusively towards either the plus-end or the minus-end. Even some motors that display oscillatory movements in both directions *in vitro* [4] undergo persistent unidirectional motion *in vivo* [5].

Unidirectional Motion

The microtubule motor kinesin I can serve as a paradigm for motor-driven transport (Figure 1). In this protein, the two motor domains responsible for motility and force generation are connected by a stalk to a globular tail, which directly or indirectly binds to cargo. The motor domains bind the microtubule tracks and undergo cyclic conformational changes driven by ATP hydrolysis which propel the kinesin molecule, and thus the attached cargo, towards the microtubule plus end (reviewed in [6,7]).

This cartoon view suggests two crucial points of regulation. Transport could be controlled at the motor domains, by altering whether motors bind to the tracks and how quickly the motors proceed through their mechanochemical cycle. Alternatively, motion could be regulated by determining whether motor molecules dock to their cargoes. The physical properties of kinesin I are understood in great detail, and the investigation of other motors has followed suit [5,8,9]. In recent years, great strides have also been made towards understanding docking; receptors on cargoes have been identified, as well as the motor subunits they interact with (reviewed in [10,11]).

In principle, the regulation of docking and of motor mechanics could control key properties of transport. The overall speed of transport depends on the fraction of time motors are docked to cargo and the rate at which they proceed through their mechanochemical cycle. Docking may control both the timing of transport, by establishing and releasing the motor–cargo link on a

Box 1

Important players in bidirectional transport.

Kinesin I: A plus-end directed microtubule motor, also known as 'conventional kinesin'. It is important for many different transport processes and has in particular been linked to several neurodegenerative diseases.

Kinesin II: A plus-end directed microtubule motor, also known as "heterotrimeric kinesin". It has important functions in neurons, intraflagellar transport and photoreceptor cells. It has been shown to cooperate with cytoplasmic dynein in pigment granule transport in frog melanophores.

Myosin V: Actin-based motor with various transport functions, including the transfer of pigment granules from microtubules onto actin filaments.

Cytoplasmic dynein: A minus-end directed microtubule motor, with roles in a wide range of cellular processes, from axonal transport to mitosis to nuclear positioning and vesicle transport. A single motor complex contains heavy chains, intermediate chains, light-intermediate chains and light chains from three distinct families. Subunit variation is thought to be important for transport specificity.

Dynactin: Multisubunit dynein cofactor that has both microtubule- and cargo-binding capabilities. It can mediate attachment of dynein to cargo and increase dynein's processivity. Dynactin is implicated in coordination of plus-end and minus-end motors in bidirectional transport.

p150/Glued: Subunit of dynactin that can physically interact with microtubules, the intermediate chain of cytoplasmic dynein and a subunit of kinesin II. Dominant-negative versions of p150/Glued are used to interfere with dynactin function.

p50/dynamitin: Another subunit of dynactin. Overexpression of dynamitin disrupts the dynactin complex and has been used as a tool to experimentally interfere with cytoplasmic dynein function. Such overexpression also abolishes plus-end motion for several cases of bidirectional transport.

Klar: Klarsicht, a novel protein. Klar is important for proper regulation of several dynein-dependent transport processes in *Drosophila*. It appears to be necessary for proper coordination of plus- and minus-end motors on lipid droplets.

PKA: Protein kinase A, otherwise known as cAMP-dependent protein kinase. cAMP binding to PKA's regulatory subunit causes release and activation of the catalytic subunit. PKA is a key component in many signal transduction pathways and mediates regulation of bidirectional transport of pigment granules and viruses.

particular schedule, and the direction of transport, by controlling whether the cargo attaches to a plus-end or minus-end motor.

Bidirectional Motion

Although initially just viewed as a curiosity, the 'saltatory' motion of cellular organelles [12–14] — where the moving organelle starts, stops and changes direction — is now recognized as a widespread phenomenon. Cargoes as diverse as secretory vesicles, mitochondria, and virus particles have been shown to move bidirectionally. Such cargoes employ motors of opposite polarity in rapid succession. Although these cargoes constantly reverse their direction of motion, they can achieve polarized distributions within the cell, by biasing the time spent moving in the plus or minus direction.

For this review, bidirectional cargoes are defined as those that move along polarized microtubules using sets of opposing motors. This definition excludes cases

in which a single type of motor switches back and forth between anti-parallel tracks, as may be the case in proximal dendrites [15]. The transport of particles inside flagella — commonly referred to as IFT for intraflagellar transport — may fall under this definition of bidirectionality, as cargoes switch from plus-end to minus-end directed motion when they reach the flagellar tip [16]. Whether these cargoes undergo multiple successive switches in direction is not yet clear.

Bidirectional transport is not simply a minor extension of unidirectional motion, so the rules that govern transport regulation in one case cannot be simply extrapolated to the other. For example, motors for both directions are stably attached to some cargoes (see below), so that establishing the connection between motors and cargoes is not the major means of regulation. Instead, directionality and timing of motion depend on correctly tuning the activities of the opposing motors, thereby modulating, for example, the length of travel in the plus-end or minus-end direction [17–22]. In some cases, the distance a cargo travels appears not to be controlled by how long individual motors can travel along the track — their processivity — but by a higher-order switch mechanism [23]. For bidirectional motion, therefore, the key challenge is to identify the mechanisms that control the relative activity of the opposing motors, a pursuit that has just begun.

Bidirectional Transport Is Common

The number of cargoes known to display bidirectional motion has steadily increased over the last few years. Among them are mitochondria [22,24], endosomes [25–27], phagosomes [28], secretory vesicles [29–31], viruses [20,21,32], intermediate filaments [33] and many identified and unidentified vesicles in neuronal axons and growth cones [34–39]. It is likely that this list will continue to grow as imaging technology improves and previously undetected reversals in motion can be resolved.

Mitochondria

In a neuron, mitochondria are present throughout the axon, but they accumulate where the need for ATP production is especially high (Figure 2). During axonal outgrowth, mitochondria show net transport towards active growth cones and away from silent ones. The net direction of transport is controlled by altering the frequency with which the mitochondria engage in plus-end or minus-end travel [40]. Growth factor signaling can regulate the net directionality: local stimulation of axons with nerve growth factor (NGF) causes accumulation of mitochondria without altering the gross behavior of other axonal cargoes [24]. Several lines of evidence suggest that phosphatidylinositols mediate this regulation [24,41].

Pigment Granules

In many vertebrates, the color of skin and hairs depends on pigment granules. In mammals, these granules are produced in melanocytes, transported to the cell periphery, exocytosed and then taken up by neighboring keratinocytes. Transport to the periphery in melanocytes is achieved by bidirectional motion along radially oriented microtubules [42]. In the periphery, the

actin-based motor myosin V actively removes the granules from their tracks.

Frogs and fish have specialized pigment cells, melanophores (Figure 3), in which pigment granules change their intracellular location in response to hormonal stimuli [43]. Granule redistribution allows the organism to adapt its skin color rapidly to the environment. When granule motion along the actin cytoskeleton is experimentally abolished, granules move bidirectionally along microtubules, driven outward, towards the plus ends, by kinesins and inward, towards the minus ends, by cytoplasmic dynein. In response to hormones, the length of runs between reversals is modulated [17,18].

The following discussion applies to fish melanophores; the situation for frogs is subtly different, based on the exact contribution of actin-based motion [17,18,44,45]. During 'aggregation', minus-end directed motion dominates, and granules accumulate in the center. During 'dispersion', plus-end directed motion dominates, and granules are found in the periphery if actin-based motion is inhibited. If actin-based motion is intact, granules transfer to the actin cytoskeleton during dispersion and spread out throughout the cytoplasm. Hormonal stimulation alters intracellular cAMP levels, which in turn modulate the activity of the cAMP-dependent protein kinase A (PKA) and ultimately alter the balance of plus- and minus-end motion [43]. Signaling through protein kinase C (PKC) has an ancillary role [46].

Lipid Droplets

Lipid droplets are storage organelles for neutral lipids, such as triacylglycerols and steroid esters. In *Drosophila* embryos, lipid droplets show developmentally regulated bidirectional transport (Figure 4). A period of balanced motion is followed by net plus-end transport; one hour later, net minus-end transport is initiated [19]. Developmental regulation involves at least two signals: one is independent of transcription, and the other involves new expression of Halo (Figure 4B), a small basic protein of unknown biochemical function [47,48]. Lipid droplets also move bidirectionally in mammalian tissue-culture cells [26].

Viruses

Many viruses hijack microtubule motors to travel quickly through the cytoplasm. Direct labeling of viral capsids has made it possible to observe this motion of viruses in real time. In infected cells, adenoviruses, for example, move bidirectionally along microtubules, with frequent reversals [20]. Infection activates two cellular signaling pathways acting through PKA and MAP kinases, respectively [49]. These signals increase both the velocity and frequency of minus-end motion and thus promote net minus-end transport towards the nucleus. It is not known which part of the motor machinery these signals act upon.

Newly assembled Herpes simplex virus (HSV) particles can travel in an infected neuron from the cell body to the axon terminals. Pseudorabiesvirus capsids move bidirectionally through the axon, with frequent reversals and longer travels towards than away from the terminals [21]. HSV particles can also

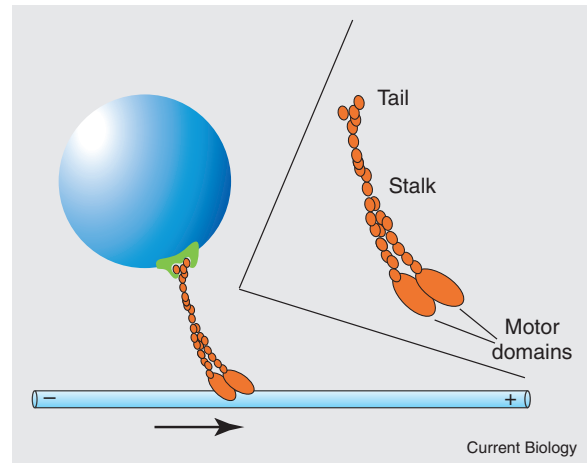


Figure 1. Unidirectional transport.

A prototypical motor with two motor domains, a stalk, and a tail domain moves cargo unidirectionally towards the microtubule plus end. The motor domains interact with the track, and the tail domain attaches to the cargo via a receptor/adaptor (green).

move bidirectionally [50], and physically interact with both dynein [51] and kinesin [52].

HIV particles in an infected cell use microtubules and cytoplasmic dynein to accumulate near the nucleus. Individual particles move back and forth between the periphery and the cell center, reminiscent of bidirectional transport [32]. For influenza viruses taken up in endosomes, an initial actin-driven motion is followed by unidirectional, dynein-dependent travel to the vicinity of the nucleus, and then microtubule-based bidirectional transport [53].

Intermediate Filaments

Intermediate filaments are the least dynamic component of the cytoskeleton. Nevertheless, under some conditions, these filaments and their precursors can move along both actin and microtubule tracks [54]. In fibroblasts, dots of GFP-vimentin move bidirectionally along microtubules. Kinesin I and cytoplasmic dynein are both required for this rapid motion and colocalize with vimentin [55,56].

Neurofilaments are synthesized in neuronal cell bodies and then move down axons by slow axonal transport (~0.3–8 mm per day) [57]. Individual filaments, however, can move short distances at speeds typical of motor-driven transport (>1 μ m per second) [58]. Net transport is slow because of frequent pausing and occasional reversals in directions. Both *in vivo* and *in vitro*, neurofilaments are found to be physically associated with kinesins and cytoplasmic dynein [33,54,59,60]. The frequent pauses might reflect either detachment of the filaments from the microtubule tracks or instances of tug-of-war when opposing motors pull on the same cargo.

Ribonucleoprotein Granules

Certain mRNAs actively move along microtubules [61], and in several instances there is evidence for bidirectional motion. In rat hippocampal neurons, a GFP-fusion of the RNA-binding protein Staufen displays fast,

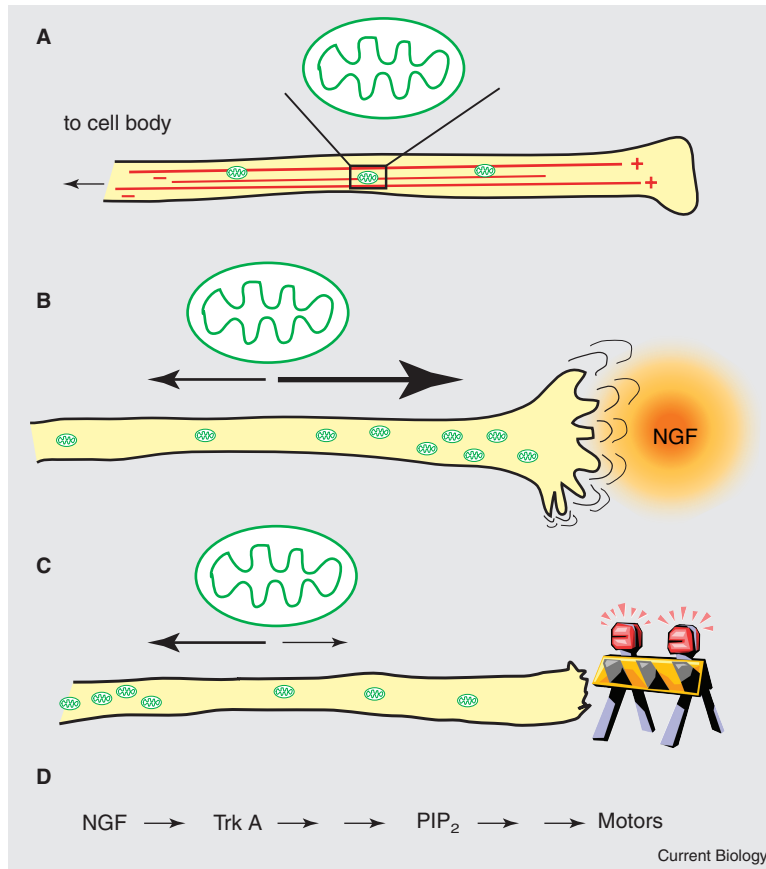


Figure 2. Transport of mitochondria in axons.

Bidirectional motion of mitochondria in axons depends on the state of the growth cone. (A) In axons, microtubules (red) are arranged with their plus ends towards the synapses or growth cones (right) and with their minus ends towards the cell body (left). Mitochondria travel bidirectionally along these microtubules. (B) When growth cones are actively growing (e.g. due to NGF stimulation), plus-end motion of mitochondria dominates over minus-end motion, resulting in net transport of mitochondria towards the plus ends. (C) If growth cones are arrested, plus-end motion is downregulated, while minus-end motion remains unaltered, resulting in net transport back to the cell body. (D) Schematic representation of motor regulation: NGF signaling through the Trk A receptor leads to production of phosphatidylinositols which, by as yet unknown mechanisms, alter the balance of motor activity.

microtubule-dependent motion, characterized by frequent pausing and occasional reversals [62]. Staufen particles indeed associate with both kinesin and dynein, as shown by immunolocalization and biochemical purification [63]. In oligodendrocytes, exogenous mRNA for myelin basic protein (MBP) assembles into granules that move bidirectionally and colocalize with both kinesin and cytoplasmic dynein (cited in [64]). Intriguingly, the localization of several mRNAs in *Drosophila* oocytes requires the activity of cytoplasmic dynein and kinesin I [65–67]. It is possible that one motor is required for correct intracellular localization of the other motor or that some of these RNAs move bidirectionally along microtubules [68]. Recent advances in imaging technology to monitor oocyte mRNA transport in real time [69,70] should make it possible to determine which of these models is correct.

Is Your Favorite Transport Process Bidirectional?

Although regulation of transport is fundamentally different for unidirectional versus bidirectional motion, distinguishing between the two types is not trivial. For some bidirectional cargoes, the balance of motion is greatly skewed, so that motion in direction A dominates and excursions in direction B last only for a few hundred milliseconds and cover a few hundred nanometers [17]. Movies taken at the rate of one image per second — reported rates of acquisition with the confocal scanning microscope, for example, are often much lower — would not reveal significant motion in

direction B, and experimenters would likely conclude that transport is unidirectional. Even if image acquisition rates are high, travel in direction B could be obscured by limited spatial resolution. On the other hand, the simultaneous presence of motors with opposite-polarity on the same cargo might imply bidirectional transport, but it could also be an instance of a temporarily inactive motor hitching a ride on a cargo moving in the opposite direction.

Few transport processes have yet been examined at high enough spatial and temporal resolution to determine confidently whether they are unidirectional or bidirectional. Once the molecular mechanisms that allow cargoes to move bidirectionally are understood, molecular markers specific for the two classes of transport may emerge. This may, for example, be important to combat novel viruses that travel along the microtubule cytoskeleton, as strategies to interfere with such trafficking are likely to depend on whether the motion is unidirectional or bidirectional. For uni-directional transport, one might target the motor or the motor-cargo link to disrupt trafficking. For bi-directional transport, it may be sufficient to just slightly skew the balance of plus-end and minus-end motion to prevent an infecting viral particle from efficiently reaching the nucleus.

Mechanisms of Bidirectional Transport

Taking advantage of a range of experimental systems, researchers have in recent years established a rough first draft of the principles that govern motor interaction

and transport regulation for bidirectional transport. The data are still too spotty to discern to what extent these insights will apply to different transport processes, but they help define the issues that will have to be addressed in the future.

How Do Multiple Motors Interact?

During unidirectional transport, motors have to be attached simultaneously to both cargo and track for motion to occur. With two opposing motors in play, three different scenarios can be envisioned, according to who is attached to whom (Figure 5). In the first scenario, only one type of motor is bound to the cargo at any one moment, possibly because the two motors share a single binding site. Reversals occur when motor Y docks and displaces motor X from the cargo. In the second scenario, both motors are simultaneously attached to both cargo and microtubules, engaging in a 'tug-of-war', with the stronger motor determining the direction of motion at any particular moment. In the third scenario, both motors are simultaneously bound to the cargo, but their activities are coordinated, so that at any given time only one motor is engaged with the track.

The first scenario (Figure 5A) can be ruled out in several systems: When in *Dictyostelium* the endogenous intermediate chain of cytoplasmic dynein is replaced with a GFP-fusion construct, fluorescent dots move along microtubules in both directions and frequently reverse course [71]. As these dots presumably represent dynein complexes attached to cargoes, dynein remains present on its cargo during plus-end motion. In addition, the bidirectional motion of frog pigment granules and mammalian endosomes can be reconstituted with purified organelles without any additional cellular proteins [25,72]. Thus, motors could not be newly recruited when the cargoes switch direction and must, therefore, be simultaneously and stably present on the cargo. This appears to be true also *in vivo*, because the total amount of either plus- or minus-end directed motors that is attached to mitochondria or pigment granules does not change even when the relative time spent in plus- versus minus-end excursions changes drastically [18,41]. Finally, mutations in the minus-end motor can alter the motion of *Drosophila* lipid droplets during plus-end travel, implying that the minus-end motor is attached to the cargo during that period [73].

If motor Y were present on the cargo and actively engaged with the microtubule during travel in the direction of motor X, as in the second scenario (Figure 5B), then impairing motor Y should improve motion in the direction of motor X. In particular, the net force for travel in the direction of motor X should increase, because motor Y will compete less effectively. This prediction was tested using *Drosophila* lipid droplets [73]. Specific mutations that impair the minus-end motor dynein or its cofactor dynactin do not result in an increase but in a decrease of plus-end directed forces, suggesting that these mutations induced a tug-of-war not seen in the wild type. Supporting evidence comes from several studies that found that treatments designed to abolish minus-end motion of organelles can also stop their plus-end travel [26,38,74,75] and vice versa [76–78]. Taken together, these observations

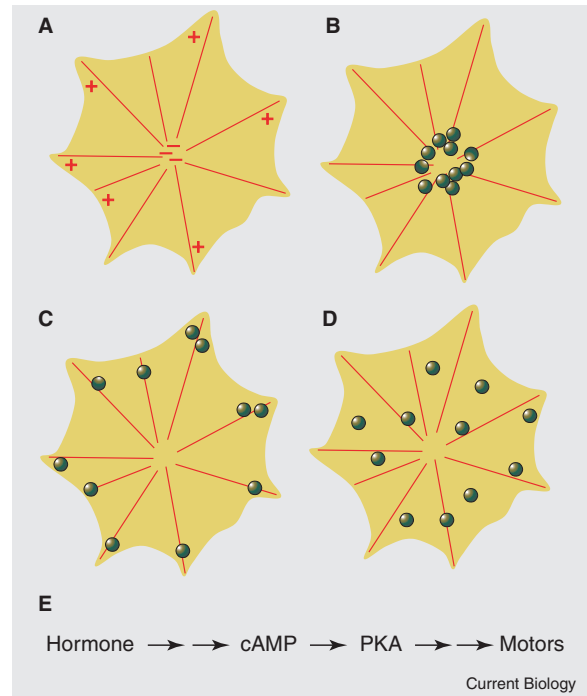


Figure 3. Pigment granules in fish melanophores.

Pigment-granule distribution is controlled by the cooperation of several motors: the minus-end directed motor is cytoplasmic dynein, the plus-end directed motor is a member of the kinesin family, and myosin V powers motion along actin filaments. (A) Microtubules are arranged radially, with minus ends in the center and plus ends towards the periphery. Actin filaments have been omitted for clarity; they are arranged in random orientation throughout the cytoplasm. (B) During aggregation, minus-end directed motion dominates and granules concentrate near the microtubule organizing center. (C) During dispersion, plus-end microtubule motion dominates and, if actin-based motion is inhibited, granules accumulate at the periphery. (D) If actin-based motion is intact during dispersion, granules frequently transfer between actin filaments and microtubules and spread throughout the cytoplasm. (E) Schematic representation of motor regulation: hormones bind to surface receptors and control the intracellular level of cAMP, which in turn regulates PKA activity. PKA directly or indirectly controls the motors.

favor a model in which motors with opposite-polarity are coordinated, as in the third scenario, rather than engaged in a tug-of-war. However, there is as yet no direct, unequivocal evidence for coordination; it is rather that the alternative scenarios appear very unlikely. A crucial challenge for future investigations will be to develop ways to monitor when motors are on the cargo, but not engaged with the track, preferably in real time, so that the status of opposing motors can be simultaneously determined.

Motor Coordination

If motors are indeed coordinated, can we discern anything about the underlying mechanisms from the data available so far? Coordination requires that the activity states of the two motors be reciprocally controlled. There have to be mechanisms that sense the activities of the opposing motors, integrate this information, and appropriately activate or inactivate one or the other

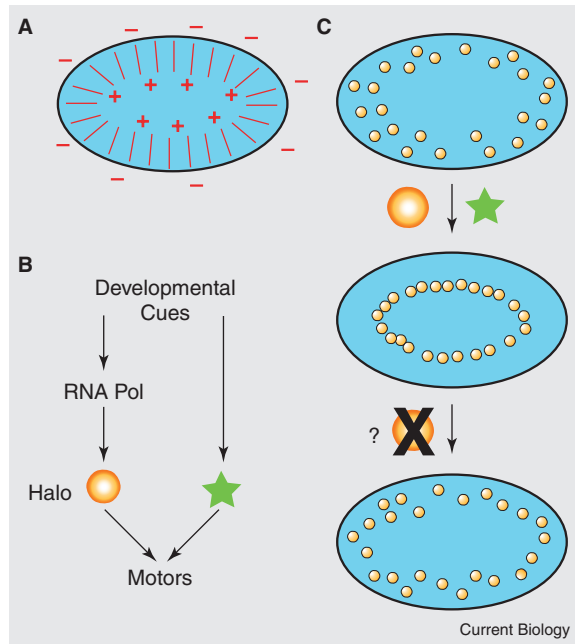


Figure 4. Lipid droplets in early *Drosophila* embryos. Lipid droplets in early *Drosophila* embryos undergo developmentally regulated redistribution. The minus-end motor is cytoplasmic dynein, the plus-end directed motor is unknown. (A) Microtubules (red) are arranged radially in the early embryo, which at this point in development is a single cell. Minus ends point to the periphery; plus ends point into the interior. Microtubules are not drawn to scale: The embryo is about 500 μm long, and the microtubules penetrate into the outer $\sim 35 \mu\text{m}$ region. (B) Schematic representation of motor regulation: Unidentified developmental cues promote the transcription of the *halo* gene and also the generation of a second, transcription-independent signal. Halo protein and the second signal cooperate to alter the balance of motor activity. (C) Distribution of lipid droplets changes during development: originally droplets are found throughout the periphery; then they undergo net inward transport and accumulate near microtubule plus ends. This transition requires both Halo and the second signal. Later, net transport is minus-end directed, and droplets are displaced outward. What mechanisms promote the switch to outward transport is unknown, though downregulation of Halo may play a role [47].

motor. Such inactivation might involve steric constraints (pushing a motor domain physically away from the track, for example), biochemical means (such as phosphorylation that prevents binding of the motor domain to the track) or both.

Information transfer between motors is most easily envisaged if motors are in direct physical contact. Although such multi-motor complexes have not yet been found, there are intriguing hints of potential physical interactions of subunits or cofactors of plus- and minus-end motors [76,79]. Even if they are unstable in free solution, such complexes might be stabilized once they are assembled in the correct geometry on the cargo surface.

Yet alternative mechanisms of coordination are possible. For example, if a membrane-anchored kinase can turn one of the motors on and the opposing motor off, then it could coordinately regulate both motors, even if they are not assembled into a single complex — as

long as the kinase can freely diffuse on the cargo surface. Alternatively, if motors turn off upon additional strain from competing motors, which is transmitted through their common attachment to the cargo, then tug-of-war states could be avoided. An analogous mechanism was suggested for the coordinated interaction at a distance of non-processive kinesins during *in vitro* microtubule gliding [80].

Mediators of Coordination

A leading candidate for a motor coordinator is dynactin, a cofactor of the minus-end motor dynein. A dominant-negative form of its p150/Glued subunit can interfere with various biological processes (e.g. [75,81,82]) and in particular reduces the force generated during lipid droplet plus-end directed motion, possibly by inducing a partial tug-of-war [73]. This is consistent with the notion that wild-type dynactin usually turns the minus-end motor off when the plus-end motor is active. No reciprocal effect on minus-end forces was observed. p150/Glued can physically interact with both the minus- and plus-end motor on frog melanosomes and the motors compete for binding to the same region of p150/Glued [76]. One of several possible models proposes that contact with dynactin stabilizes a motor's interaction with the track, thus turning it on [47,83]. If dynactin alternately contacts the two motors, then only one motor can be in the on-state at any given time. Overexpression of the p50/dynamitin subunit of dynactin disrupts the dynactin complex [84] and abolishes both plus- and minus-end motion of several bidirectionally moving cargoes [26,76], further supporting a crucial role for dynactin in motor coordination.

Reciprocal coordination of motors on *Drosophila* lipid droplets may be mediated by the Klar protein. In its absence, droplet motion displays severely impaired velocity, stall forces, and mobility perpendicular to the travel direction, but both sets of motors are attached and functional, as droplets still move bidirectionally [19]. It was proposed that without Klar motors engage in a severe tug-of-war [19]. Klar's sequence is almost entirely novel, and no clear-cut Klar homolog exists outside the insects [85]. Thus, other proteins must play similar coordination roles in other species.

Klar is also important for the migration of photoreceptor nuclei [19,86] and for vesicle delivery in developing salivary glands [87]. A tagged Klar construct expressed in photoreceptors is present around the nucleus and this localization is disrupted in lamin mutants [88]. It was proposed that lamin connects directly or indirectly to Klar, which in turn anchors dynein to the nuclear envelope [88]. Because we do not yet know whether these nuclei move uni- or bidirectionally, an alternative possibility is that Klar coordinates dynein and a plus-end motor present on the nuclear envelope. Intriguingly, these nuclei switch their overall direction of migration during normal development [89,90] and migrate basally instead of apically in the absence of Klar [19,86]. Plus- and minus-end motors are both necessary for migration of nuclei in another tissue, the *Drosophila* oocyte [65,66], but it is not clear whether that indicates that these nuclei move bidirectionally [68].

Features of Coordination

The proteins that mediate coordination seem a likely target for the action of *trans*-acting regulators. For example, if such regulators control the duration the coordination machinery spends in the minus- or plus-end state (Figure 5C), then they control lengths of runs in a given direction and thus ultimately the net direction of transport (see below). As a consequence, to understand when and where transport occurs, it may be more important to unravel the properties of the as-of-yet hypothetical coordination machinery than the properties of the motors in isolation.

Because the behavior of the two opposing motors is coupled in a non-trivial manner, experimental manipulation of one motor can have complicated effects on motion in the opposite direction. In *Xenopus* melanophores, for example, overexpression of the plus-end motor subunit that is implicated in binding to the coordination machinery as well as the disruption of dynein impair transport in both directions [76]. As a corollary, if one experimentally interferes with a particular motor, the observed phenotypes could be due to disruption of coordination and alteration of motion in the opposite direction. It is, therefore, paramount to determine if transport is uni- or bidirectional before such manipulations can be fully interpreted.

A case in point are the intriguing functional interactions of kinesin I and cytoplasmic dynein during axonal transport. Disruption of either motor causes transport defects for both directions, including stalled cargoes and axonal swelling [75,91]. Mutations in both motors cause markedly stronger defects [75]. In contrast, reduction of dynein levels can ameliorate defects that are due to overexpression of the kinesin receptor APP, while kinesin I reduction enhances such defects [2]. In some cases, bidirectional transport may be disrupted due to breakdown of coordination, while in others roadblocks caused by one motor may indirectly interfere with transport in the opposite direction [75]. Deciphering the underlying mechanisms could have profound impacts on the treatment of neurodegenerative diseases. Such an analysis will presumably require tracking known cargoes in axons at high resolution to determine the extent of bidirectional transport and to quantify the effects of motor mutants.

Finally, overexpression of the dynactin subunit p50/dynamitin physically disrupts the dynactin complex [84] and has become an invaluable tool for interfering with dynactin function *in vivo*. However, the consequences of such overexpression may not be simply due to the inhibition of cytoplasmic dynein, as is often concluded, but also due to interfering with plus-end motion of a bidirectionally moving cargo, especially as dynactin appears to play a crucial role in motor coordination [83].

How is Directionality Determined?

At least two general models can be envisioned how bidirectionally moving cargoes achieve net transport. Regulation could either affect the bias between plus- and minus-end motion or the loading and removal of cargoes at the ends of the tracks.

If cargoes fall off the microtubule track after reaching one particular end and are then sequestered, this end

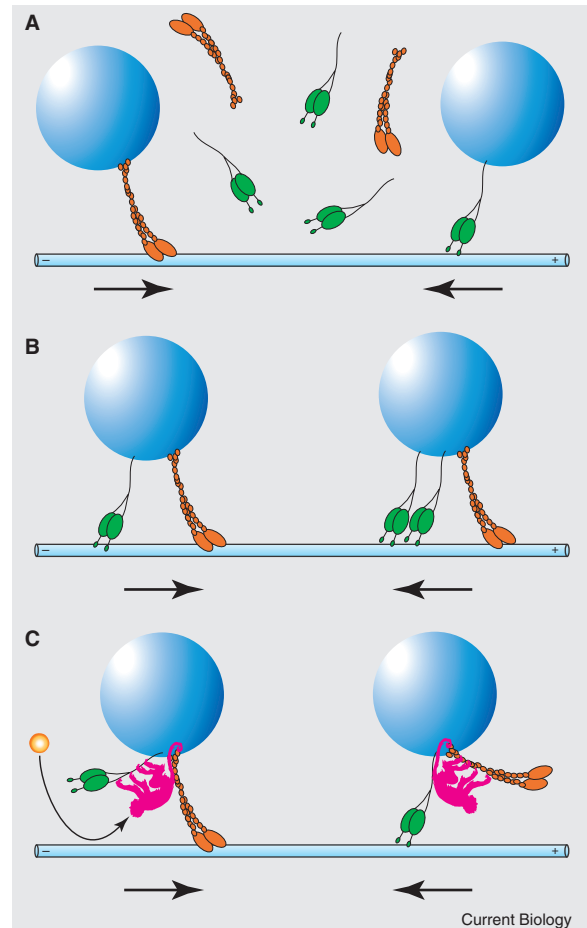


Figure 5. Three possible mechanisms of bidirectional transport. Plus-end motors are shown in orange, minus-end motors in green. Cargo receptors have been omitted for simplicity. (A) At any moment, only one type of motor is attached to the cargo. Reversals come about when this motor is replaced by a motor with opposite polarity. (B) Motors for both directions are always attached to both cargo and microtubules, working against each other. The direction of cargo movement depends on the relative strength of the competing motors. In the example shown, one plus-end motor is stronger than a single minus-end motor (left) but weaker than two (right). Thus, the cargo on the left moves towards the plus end and the cargo on the right towards the minus end. Alteration in motor number is only one way how the power for motion in a given direction might be modulated, as it is possible to vary the force produced by a single motor [8]. (C) Opposing motors are simultaneously attached to the cargo, but a complex coordination machinery (pink) ensures that when one motor is actively engaged with the microtubule, the other motor is turned off [47,83]. The coordination machinery might be the target of *trans*-acting regulators. If the coordination machinery can attach to cargo independent of the motors, distinct variants of the coordination machinery could be targeted to different cargoes, thus allowing cargo-specific coordination and regulation.

acts as a sink for transport. Bidirectional motion would ensure that cargoes continue to be delivered to the sink. Such a sink mechanism has been proposed to operate for pigment granules in mammalian melanocytes, where granules move bidirectionally along radial microtubules and get captured by myosin V when they reach the actin-rich cell cortex [42].

If, on average, plus-end moving cargoes travel a distance p before reversing course, but minus-end moving cargoes travel a distance m , then the direction of net transport should be towards the plus end if $p > m$ and towards the minus end if $p < m$. Such a biased random walk underlies the net transport of fish and frog pigment granules, embryonic lipid droplets, adeno- and herpesviruses, and axonal mitochondria [17–22]. Although the difference ($p - m$) gives a qualitative measure of net directionality, the exact rate of transport depends also on the speed of runs, the time from reversal to reversal, and the frequency of pauses [19,92,93].

The distance traveled between reversals is, therefore, a key parameter for understanding transport, and I will refer to the motion from one reversal to the next as a ‘trip’. The more euphonious term ‘run’ already has an accepted definition as a period of uninterrupted motion. As a cargo travels from one reversal point to the next, it might pause many times — so a trip can be made up of many runs. If pausing is rare, trips and runs will be essentially the same. Whether runs or trips are the more important terms to gain mechanistic insight will probably depend on the nature of pauses: whether they are crucial transitions between runs of distinct properties or temporary interruptions of trips with similar features before and after the pause. Most published reports have described runs instead of trips. In the discussion below of how travel distances are regulated, I shall therefore almost exclusively talk about runs, though for net transport the control of trips is most relevant.

Run Length Control

The distribution of run lengths provides a first hint about the underlying control mechanisms. If there were a mechanism to end runs after a certain number of steps, then run lengths should cluster around a particular value. In the systems in which run length distributions have been reported so far, however, run lengths display a monotonically decaying distribution, so that very short runs are frequent and longer runs are increasingly rare [18,21,23]. These distributions resemble the exponential decay seen for run lengths of motors *in vitro*, in which the motor has at every step a constant probability of falling off the track and ending the run. Thus, run length might simply reflect the inherent processivity of the motor (i.e. the number of steps motors can take without falling off the track), and run length control might involve regulating motor processivity. An underappreciated twist is that in some cases run lengths are best described as the sum of two decaying exponential distributions [18,23], possibly indicating that runs are driven by motors in at least two distinct biochemical states.

Several observations indicate that motor processivity is not the crucial limiting factor for runs *in vivo*. Many runs are much shorter than one would expect from the processivity of motors *in vitro* [18, 23], and reversals of travel direction can be extremely rapid (e.g. less than 33 ms for lipid droplets [23]); if runs were ended by motors falling off the tracks, runs would be followed by pauses or diffusion before motion resumed. Such diffusive pausing has indeed

been observed during axonal transport for cargoes that display little or no tendency to move bidirectionally [94]. Thus, *in vivo* run lengths may be substantially shorter than inherent motor processivity, and control of run length must then involve actively cutting runs short. Run length control might be achieved by simply flipping the hypothesized coordination machinery from a plus- to a minus-end state, and vice versa (Figure 5C).

Run Length Control via Multiple Motors?

A simple way to control run length would seem to be changing the number of active motors. For example, if a motor goes, on average, 100 steps before falling off — in other words, at each step, the probability of ending a run is 1% — then two motors could go some 10,000 steps if motors stay on the cargo and can reengage as long as the cargo remains near the track. Cells seem to employ such a mechanism to control vesicle transport by the motor Kif1A [95]. Monomeric Kif1A tends to fall off the tracks frequently. When individual motors are in close proximity on a single cargo, Kif1A dimers with a much higher processivity form, presumably because one motor domain can hold the motor at the track and prevent diffusion while the other motor steps [95].

There is little information about how many active motors a cargo carries *in vivo*. For *Drosophila* embryonic lipid droplets, stall forces change during development in a quantized manner [19]. This was interpreted as a change in the number of actively engaged motors per cargo, and the stall force for a single cytoplasmic dynein was predicted to be 1.1 pN [23], remarkably close to the maximal stall force for dynein as measured *in vitro* [8]. Yet these multiple motors per cargo do not seem to control run length: runs are much shorter than expected for multiple dyneins, and their length does not change as forces change [23].

But if multiple motors are not used for run length control, why are they there? Multiple motors may ensure that motor processivity is never limiting and might also allow big cargoes to efficiently push through cytoskeletal networks of the cytoplasm.

Biochemical Mechanisms

In several systems, protein kinases have crucial roles in determining directionality. For fish and frog pigment granules, PKA transduces extracellular stimuli and promotes granule accumulation near the nucleus by up-regulating minus-end trip lengths while only slightly affecting plus-end trip lengths (Figure 3E) [17,18]. For infecting adenoviruses, PKA and p38/MAP kinase cooperate to upregulate the velocity and frequency of minus-end motion [49]. In neither case are the functionally relevant targets of these kinases known.

Two recent papers [24,41] have provided insight into how regulation of transport of mitochondria in axons occurs (Figure 2D). Mitochondria, but not other cargoes, accumulate at sites of local NGF application [24]. Inhibitor experiments suggest that the signal is transmitted through the Trk A receptor and the phosphoinositide-3 pathway [24]. When pleckstrin homology (PH) domains, which bind phosphatidylinositols, were overexpressed in a pre-neuronal cell line, the

distribution of mitochondria, but not of endosomes, was shifted towards microtubule plus ends, due to a change in the balance of plus- and minus-end travel. Travel velocities, the amount of motors on cargoes, and motor binding to tracks were not altered [41], suggesting that specifically the relative activity of the opposing motors had been altered.

In *Drosophila*, the transacting factor Halo (Figure 4B) determines the directionality of lipid-droplet transport [47]. At a certain time in development (second panel in Figure 4C), the net transport of droplets is directed towards the minus-end in the absence of Halo, and plus-end directed in the presence of Halo. With Halo present, the plus-end runs of individual droplets are longer, and minus-end runs are shorter, but the velocities are unaltered. Expression and inhibitor studies suggest that Halo acts directly or indirectly *in trans* on a pre-existing motor machinery.

In none of these cases do we understand the molecular events that alter the behavior of the motor machinery. A crucial next step will be to identify the functionally important targets of regulation. That should reveal whether these regulators act in fundamentally similar pathways or if different systems have entirely distinct run length control.

Biological Reasons for Bidirectional Transport

To simply relocate a cargo from point A to point B, unidirectional transport would appear to be a simpler and more economical choice than bidirectional transport, in terms of both speed and energy consumption [93]. Yet bidirectional transport is common and is, therefore, likely to provide distinct biological advantages. The following provides speculations as to what such advantages might be.

Economy

If cargoes always carry motors for both directions, net transport can easily be adjusted or even reversed by simply tweaking the relative activity of the two motors. This is likely to be much quicker than assembling a new set of motors on a cargo, and also allows transport to be abruptly altered depending on cellular needs. It even makes it possible to tune the overall speed of transport by altering the relative contribution of trips in the non-dominant direction.

Bidirectional transport may also shed light on the vexing problem of how cargo specificity is regulated. Cytoplasmic dynein, for example, acts in many different transport processes in the same cell, yet they all occur under their characteristic spatial and temporal control [96–99]. Non-motor proteins specifically targeted to particular cargoes might control the frequency of switching between the states in which the plus- and minus-end motors are active (Figure 5C). Thus, if regulation occurs at the level of the coordination machinery, then cargoes can employ identical motors, yet be regulated differently. Intriguingly, proposed components of the coordination machinery exist in several functional forms: neurons express a biochemically distinct form of dynactin [100,101], and distinct isoforms of Klar are targeted to different cargoes (Y. Guo and M. Welte, unpublished observations).

Setting up Polarized Distributions

Cooperation between opposite-polarity motors is crucial for positioning organelles, such as the Golgi [102,103] or the oocyte nucleus in *Drosophila* [65,66]. Sometimes it is necessary to set up a distribution rather than to confine the organelles to a single point: mitochondria in axons accumulate in areas of high ATP demand, yet some mitochondria remain in less active regions. Even if cargoes accumulate at a certain point (e.g. near plus-ends when motion is biased in the plus-end direction), trips in the non-dominant direction will tend to spread the cargoes out along the tracks, away from the point of accumulation. Modeling shows that by altering the relative contributions of plus- and minus-end trips, a wide range of steep to flat steady-state distributions can be achieved [92].

Avoiding Obstacles and Exploring Space

Microtubules are straight, hollow cylinders whose walls are made up of strings of tubulin heterodimers arranged in head-to-tail fashion. Typically 13 such protofilaments lie side-by-side around an empty lumen. Kinesin I moves along single proto-filaments [104,105]. This extraordinary precision could become a liability if another motor or a temporary obstruction blocked the path along the proto-filament. As cytoplasmic dynein often steps sidewise to adjacent proto-filaments [106], a bidirectional cargo could find itself on the opposite side of the microtubule even after a short minus-end excursion. If it now switches back to kinesin I, it can pass the obstacle. Bidirectionally moving cargoes should, therefore, be less likely to contribute to disastrous traffic jams, such as those in axons, which have been proposed as the crucial injury in a range of neurodegenerative diseases [3,91].

The random walk of bidirectional cargoes allows a single cargo to explore a large region of cellular space, especially if tracks are disordered. For vesicles on their way to another cellular compartment, such motion increases their chance to bump into their target ('trial-and-error' model of vesicle delivery [29]). Motor driven vesicle motion with negligible net displacement has indeed been observed [29,107].

Error Correction

During unidirectional transport, the critical event that determines directionality of motion is the attachment to either a plus- or minus-end motor. A wrong attachment will cause misdelivery of the cargo. During bidirectional transport, the net direction of transport is determined by the balance of plus- and minus-end trips and can, therefore, be continually evaluated and even altered if physiological conditions change. Thus, bidirectional transport may facilitate error correction.

Switching between alternative states of motion is a general strategy to improve accuracy. For instance, DNA polymerase switches between 5'-to-3' elongation and 3'-to-5' removal of incorrectly incorporated bases. Similarly, bacterial flagellar motors switch between counter-clockwise rotation resulting in sustained forward motion of the bacterium (runs) and clock-wise rotation resulting in random reorientation of the

bacterium (tumbles). This makes it possible to constantly reassess travel direction, such that initial errors and errors arising during travel can be corrected [108].

Beyond Bidirectionality

Bidirectional transport by opposite-polarity microtubule motors is just one example of multiple motors working together to achieve carefully choreographed transport. A thorough understanding of bidirectional transport should make it easier to dissect even more baroque systems in which the activities of many motors have to be balanced.

A well-studied case of cooperation between actin- and microtubule-based motors is the transport of pigment granules in fish and frogs. In *Xenopus*, pigment granules move along microtubules using cytoplasmic dynein and kinesin II, and switch to movement along actin filaments driven by myosin V. Granules switch from microtubules to actin filaments only during minus-end runs [18], possibly because myosin V can compete against cytoplasmic dynein in a tug-of-war, but cannot pull cargoes off the microtubules when the plus-end motor is active. In fish melanophores, the activity of the three motors is coordinately controlled by cAMP levels, so that at low cAMP concentrations microtubule minus-end motion predominates. At intermediate cAMP levels, myosin V activity is high. Consequently, there is a mix of plus-end and minus-end microtubule-based motion, which allows for a switching from microtubules to actin filaments. At high cAMP levels, plus-end motion is large and there is little minus-end motion such that microtubule to actin filament switching is severely reduced [17].

During mitosis, a large number of different motors have to work together to establish and maintain the spindle and to segregate chromosomes. Motors help organize the spindle poles and the spindle midzone, position the spindle relative to the cell cortex and move chromosome arms away from the spindle poles (reviewed in [109–111]). Kinetochores, in particular, contain a collection of proteins that control their motion towards and away from the poles — motors that move them relative to the microtubules and regulatory factors that control growth and shrinkage of microtubule ends. Because the behavior of kinetochores on sister-chromatids is coupled, chromosomes as a whole move bidirectionally. Thus, the behavior of several different motors in two kinetochores has to be carefully choreographed. Both information transfer by detection of tension and kinases play roles in this regulation, but the detailed mechanisms remain to be unraveled.

Open Problems

Bidirectional transport is one of the simplest examples of multi-motor machines employed in cells, but it is still far more complex than transport by individual motors. For a host of questions, we do not yet have satisfying answers: Which characteristics of transport can be derived from the properties of motors in isolation and which aspects require a higher level of

regulation? What is the mechanistic basis for coordination? How is cargo-specificity of transport established? And how is motor behavior integrated with other cellular and developmental processes? The good news is that with the basic characterization of several model systems and first inroads into the molecules involved in regulation and coordination, our understanding of bidirectional transport should improve quickly in the next few years.

The few molecules known to have crucial roles in bidirectional transport (such as dynactin and Halo) can act as starting points for identifying other components of the machinery. As several purified cargoes still move bidirectionally *in vitro* [25,72], proteomics approaches can now be used to characterize the entire set of proteins on these cargoes, a set which must include motor regulators and coordinators.

For fluorescently labeled cargoes, the rate of acquiring images is often limited by signal intensity, photobleaching or the maximal speeds of confocal scanning or cameras. Yet to identify which physical parameters are the target of regulation, high resolution tracking of cargo motion is essential. Pushing the limits of spatial and temporal resolution will therefore be crucial for extending initial insights from a few model systems to the broad range of cargoes that move bidirectionally.

The spatial arrangement of motors, coordinators, cargo, and tracks is likely to be an important piece of the puzzle of how multiple motors can be regulated in concert. It should be possible to reconstruct the geometric arrangement of these various players using electron microscopy. But in the long run it will be necessary to develop probes that detect the conformational states of motors and motor complexes *in vivo*. Once these different conformational states have been identified, it may be possible to discern the rules governing transitions between them and to develop quantitative models of how the different states interconvert. Just such an analysis has, for example, been used to model the detailed behavior of DNA polymerase [112].

We already understand a great deal about how motors function in isolation and how they dock with cargoes. To push the analysis of intracellular motion to the next higher level, it will be necessary to unravel the mechanisms that allow motors to work together in a coordinated fashion. But the rewards of this next step are likely to be great, because transport processes that require the coordination of multiple motors lie at the heart of a wide range of fundamental processes in biology.

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